

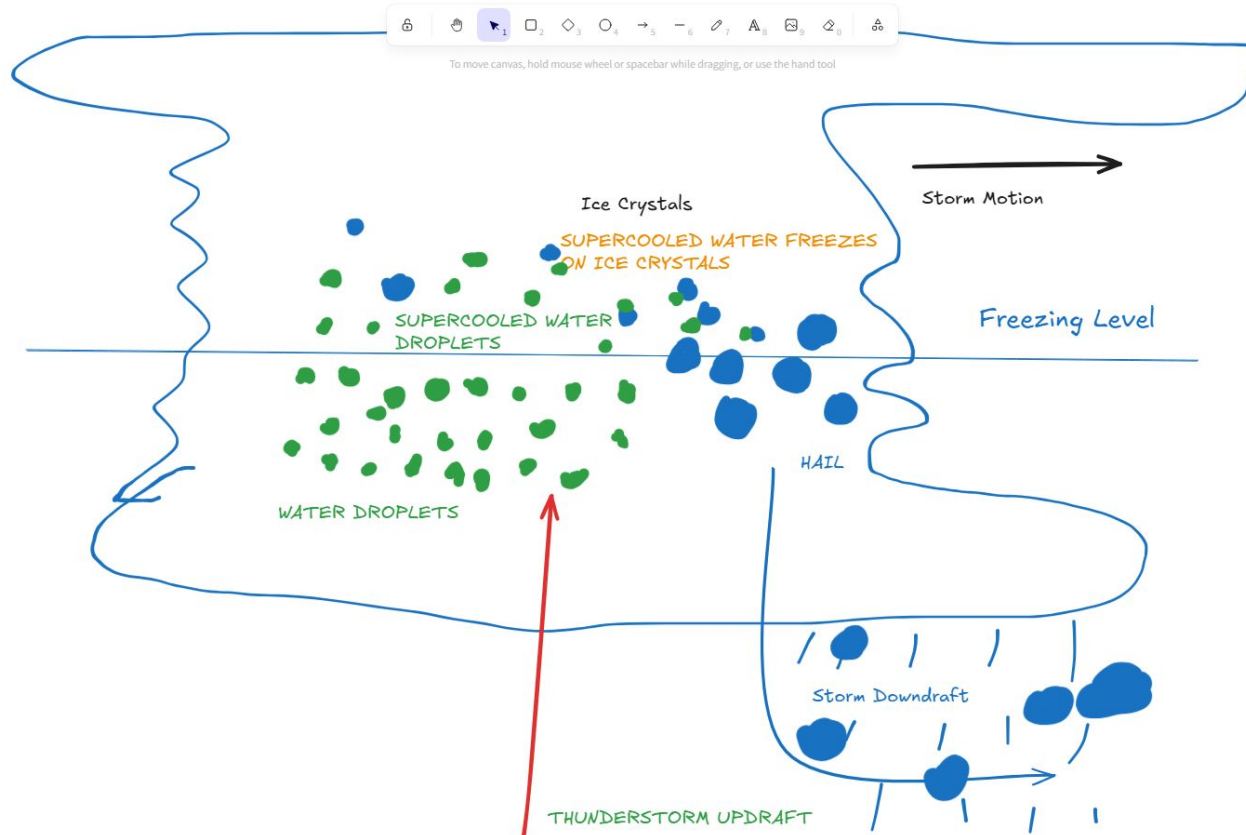
Hail vs. Snow/Sleet/Freezing Rain

Thursday, June 5, 2025

Presented By
Max's Weather Service Newport Beach CA



How Does Hail Form?

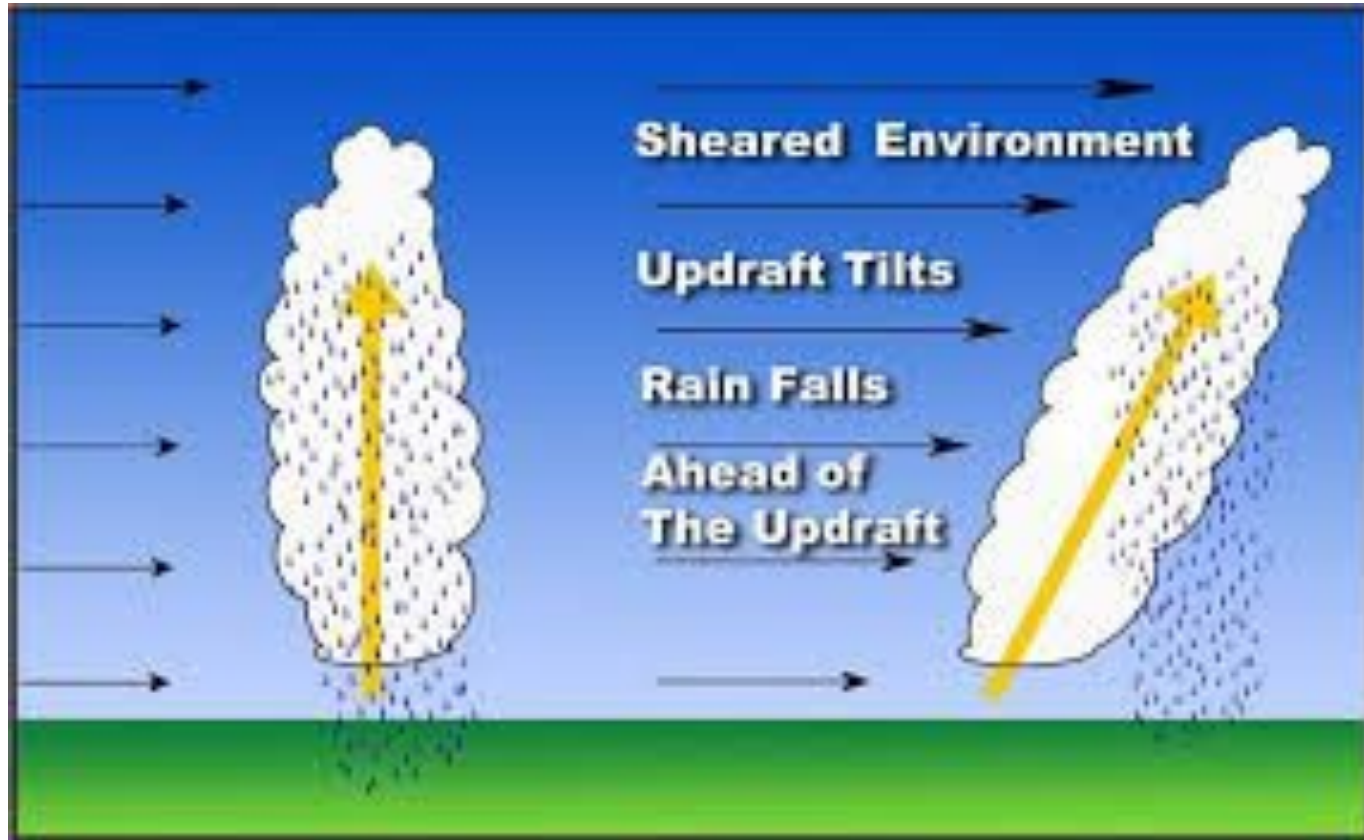


What atmospheric conditions are favorable for the formation of large hail?

1. Strong thunderstorm updraft
2. Strong deep layer wind shear (strong change in wind speed from surface to upper levels)
3. Weak low level wind shear (weak change in wind speed from surface to low levels)
4. Very unstable air mass with strong instability near the freezing level.

NOTE: The surface temperature has NO IMPACT on hail formation as long as other atmospheric conditions are present.

Deep Layer Shear



Unsheared Environment

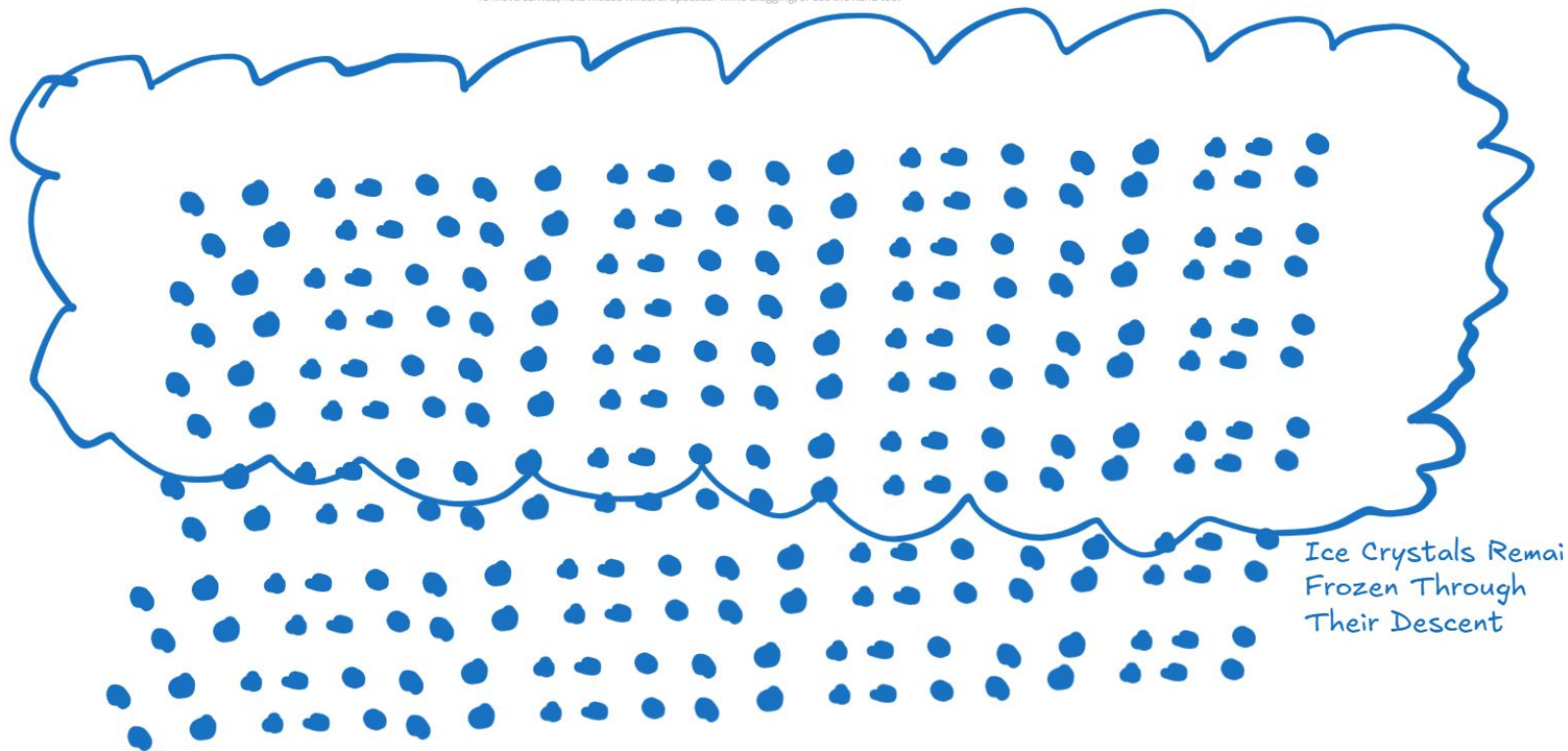
Sheared Environment

How Does Snow Form?



Share

To move canvas, hold mouse wheel or spacebar while dragging, or use the hand tool

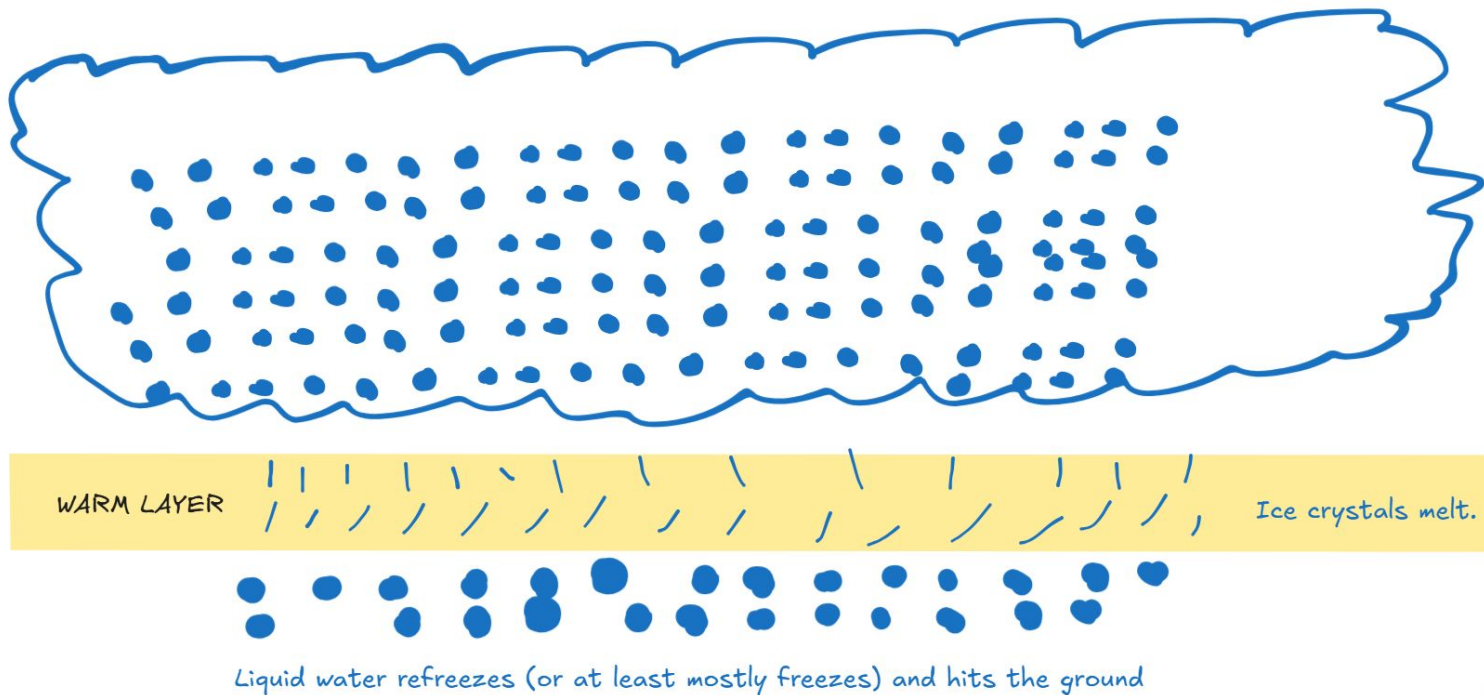


Ice Crystals Remain
Frozen Through
Their Descent

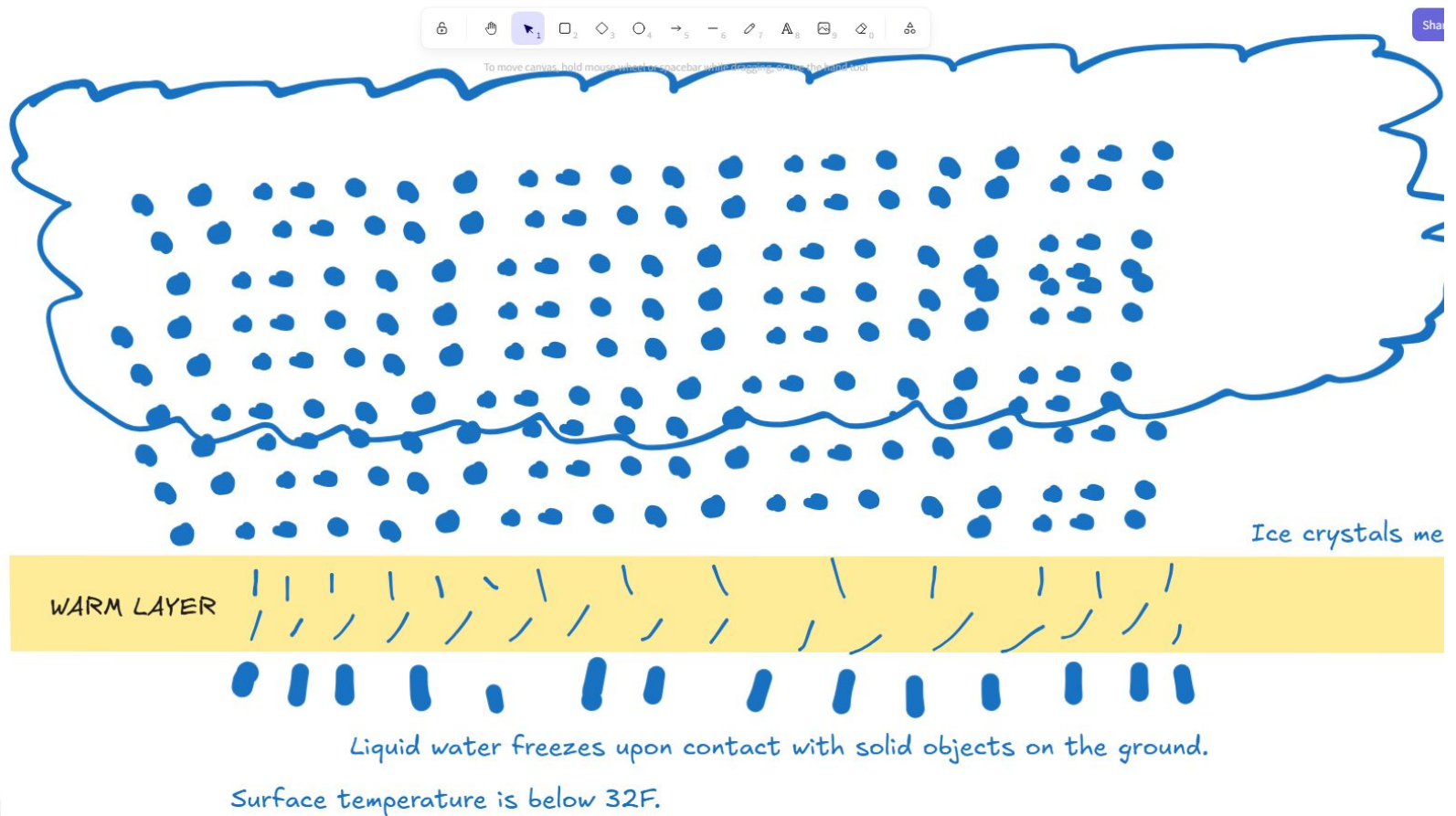
Surface Temperature is well below 32F

How Does Sleet Form?

no more carries, more mouse wheel or spacebar, better dragging, or use the touch bar



How Does Freezing Rain Form?



Thundersnow?

Wintry Precipitation-producing clouds are rarely convective.

Wintry Precipitation includes snow, sleet, and freezing rain. |

This is because convective storms require warm, moist air at the surface to work with. But, in winter weather events, the surface temperature is very cold, precluding destabilization in the atmosphere.

Even if convective storms with wintry precipitation develop (i.e. thundersnow), the chance of hail is nearly zero due to the weak instability.

This is also the reason why hail is rare in SoCal. Even with thunderstorms, instability is generally weak and therefore hail potential is low.

Convective vs. Non-Convective Precipitation Producers

CONVECTIVE

Warm-sector Supercells (with tornadoes)
Monsoon thunderstorms
Post-cold front thunderstorms
Thunderstorms along cold front (squall line)
Warm advection showers

NON-CONVECTIVE

Warm front rain (stratiform rain)
Atmospheric River Rain
Marine Layer Drizzle
Winter Precipitation